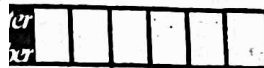




# TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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**COMMON HALF YEARLY EXAMINATION - 2024 - 25**  
**BUSINESS MATHEMATICS AND STATISTICS**

Time Allowed : 3.00 Hours]

[Max. Marks : 90

**PART - I****20 x 1 = 20**

1. Answer all the questions by choosing the correct answer from the given 4 alternatives
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark

1. The cofactor of  $-7$  in the determinant  $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$

- (a)
- $-18$
- (b)
- $18$
- (c)
- $-7$
- (d)
- $7$

2. If  $A$  and  $B$  non-singular matrix then, which of the following is incorrect?

- (a)
- $A^2 = I$
- implies
- $A^{-1} = A$
- (b)
- $A^{-1} = I$
- 
- (c) If
- $AX = B$
- then
- $X = B^{-1}A$
- (d) If
- $A$
- is square matrix of order 3 then
- $|\text{adj } A| = |A|^2$

3. The greatest positive integer which divide  $n(n+1)(n+2)(n+3)$  for all  $n \in \mathbb{N}$  is

- (a)
- $2$
- (b)
- $6$
- (c)
- $20$
- (d)
- $24$

4. There are 10 true or false questions in an examination. Then these questions can be answered in

- (a) 240 ways (b) 120 ways (c) 1024 ways (d) 100 ways

5. The x-intercept of the straight line  $3x + 2y - 1 = 0$  is

- (a)
- $3$
- (b)
- $2$
- (c)
- $\frac{1}{3}$
- (d)
- $\frac{1}{2}$

6. The equation of the circle with centre on the x axis and passing through the origin is

- (a)
- $x^2 - 2ax + y^2 = 0$
- (b)
- $y^2 - 2ay + x^2 = 0$
- (c)
- $x^2 + y^2 = a^2$
- (d)
- $x^2 - 2ay + y^2 = 0$

7. The value of  $\cos(-480^\circ)$  is

- (a)
- $\sqrt{3}$
- (b)
- $-\frac{\sqrt{3}}{2}$
- (c)
- $\frac{1}{2}$
- (d)
- $-\frac{1}{2}$

8. The value of  $\frac{3\tan 10^\circ - \tan^3 10^\circ}{1 - 3\tan^2 10^\circ}$  is

- (a)
- $\frac{1}{\sqrt{3}}$
- (b)
- $\frac{1}{2}$
- (c)
- $\frac{\sqrt{3}}{2}$
- (d)
- $\frac{1}{\sqrt{2}}$

9. The minimum value of the function  $f(x) = |x|$  is

- (a)
- $0$
- (b)
- $-1$
- (c)
- $+1$
- (d)
- $-\infty$

10. If  $y = \log x$ , then  $y_2 =$ 

- (a)
- $\frac{1}{x}$
- (b)
- $-\frac{1}{x^2}$
- (c)
- $-\frac{2}{x^2}$
- (d)
- $e^2$

11. Relationship among  $MR$ ,  $AR$  and  $\eta_d$  is

- (a)
- $\eta_d = \frac{AR}{AR-MR}$
- (b)
- $\eta_d = AR - MR$
- (c)
- $MR = AR = \eta_d$
- (d)
- $AR = \frac{MR}{\eta_d}$





12. If  $q = 1000 + 8p_1 - p_2$ , then  $\frac{\partial q}{\partial p_1}$  is

(a) -1

(b) 8

(c) 1000

(d)  $1000 - p_2$

13. If 'a' is the annual payment, 'n' is the number of periods and 'i' is compound interest for ₹ 1 then future amount of the ordinary annuity is

(a)  $A = \frac{a}{i}(1+i)[(1+i)^n - 1]$

(b)  $A = \frac{a}{i}[(1+i)^n - 1]$

(c)  $P = \frac{a}{i}$

(d)  $P = \frac{a}{i}(1+i)[1 - (1+i)^{-n}]$

14. Example of contingent annuity is

(a) Installments of payment for a plot of land

(b) An endowment fund to give scholarships to a student

(c) Personal loan from a bank

(d) All the above

15. The correct relationship among A.M., G.M. and H.M. is

(a)  $A.M. < G.M. < H.M.$

(b)  $G.M. \geq A.M. \geq H.M.$

(c)  $H.M. \geq G.M. \geq A.M.$

(d)  $A.M. \geq G.M. \geq H.M.$

16. The events A and B are independent if

(a)  $P(A \cap B) = 0$

(b)  $P(A \cap B) = P(A) \times P(B)$

(c)  $P(A \cap B) = P(A) + P(B)$

(d)  $P(A \cup B) = P(A) \times P(B)$

17. Correlation co-efficient lies between

(a) 0 to  $\infty$

(b) -1 to +1

(c) -1 to 0

(d) -1 to  $\infty$

18. The correlation coefficient

(a)  $r = \pm \sqrt{b_{xy} \times b_{yx}}$

(b)  $r = \frac{1}{b_{xy} \times b_{yx}}$

(c)  $r = b_{xy} \times b_{yx}$

(d)  $r = \pm \sqrt{\frac{1}{b_{xy} \times b_{yx}}}$

19. A solution which maximizes or minimizes the given LPP is called

(a) a solution

(b) a feasible solution

(c) an optimal solution

(d) none of these

20. Network problems have advantage in terms of project

(a) Scheduling

(b) Planning

(c) Controlling

(d) All the above

## PART - II

1. Answer any 7 questions

2. Each question carries 2 marks

3. Question number 30 is compulsory

**7 x 2 = 14**

21. The technology matrix of an economic system of two industries is  $\begin{bmatrix} 0.50 & 0.30 \\ 0.41 & 0.33 \end{bmatrix}$ . Test whether the system is viable as per Hawkins Simon conditions.

22. Find the equation of the circle having the centre (3,5) and radius 5 units

23. For the function  $y = x^3 + 19$ , find the values of x when its marginal value is equal to 27.

24. A man travelled by car for 3 days. He covered 480 km each day. On the first day he drove for 10 hours at 48 km. an hour. On the second day, he drove for 12 hours at 40 km an hour and for the last day he drove for 15 hours at 32 km. What is his average speed?

25. From a class of 32 students, 4 students are to be chosen for a competition. In how many ways can this be done?

26. Find  $y_2$  for the function :  $y = e^{3x+2}$

27. The chairman of a society wishes to award a gold medal to a student getting highest marks in Business Mathematics and Statistics. If this medal costs to ₹ 9,000 every year and the rate of compound interest is 15%, then what amount is to be deposited now.

28. From the following data calculate the correlation coefficient  $\Sigma xy = 120$ ,  $\Sigma x^2 = 90$ ,  $\Sigma y^2 = 640$ .







29. Prove that  $\frac{\sin(-\theta)\tan(90^\circ-\theta)\sec(180^\circ-\theta)}{\sin(180^\circ+\theta)\cot(360^\circ-\theta)\operatorname{cosec}(90^\circ-\theta)} = 1$

30. Find the elasticity of supply for the supply function  $x = 2p^2 + 5$  when  $p = 3$ .

## PART - III

1. Answer any 7 questions

2. Each question carries 3 marks

3. Question number 40 is compulsory

$7 \times 3 = 21$

31. If  $\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = 448$ , then find the least positive integer  $n$ .

32. If  $X = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 1 & 0 & -1 \end{bmatrix}$  and  $Y = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & n & n \end{bmatrix}$  then, find  $p, q$  if  $Y = X^{-1}$

33. If the production of a firm is given by  $P = 4LK - L^2 + K^2, L > 0, K > 0$ , Prove that  $L \frac{\partial P}{\partial L} + K \frac{\partial P}{\partial K} = 2P$ .

34. Evaluate:  $\frac{(3!)! \times 2!}{5!}$

35. If the dividend received from 9% of ₹20 shares is ₹1,620, then find the number of shares.

36. The parabola  $y^2 = kx$  passes through the point  $(4, -2)$ . Find its latus rectum and focus.

37. An unbiased die is thrown twice. Let the event A be odd number on the first throw and B the event odd number on the second throw. Check whether A and B events are independent.

38. If  $\tan(x + y) = 42$  and  $x = \tan^{-1}(2)$ , then find  $y$ .

39. Draw the event oriented network for the following data:

| Events       | 1 | 2 | 3 | 4   | 5 | 6   | 7   |
|--------------|---|---|---|-----|---|-----|-----|
| Immediate    | - | 1 | 1 | 2,3 | 3 | 4,5 | 5,6 |
| Predecessors | - | 1 | 1 | 2,3 | 3 | 4,5 | 5,6 |

40. If  $\sin y = x \sin(a + y)$ , then prove that  $\frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a}$ .

## PART - IV

1. Answer all the questions

2. Each question carries 5 marks

$7 \times 5 = 35$

41. a) Find the equation of the circle passing through the points  $(0,0)$ ,  $(1,2)$  and  $(2,0)$ .

(OR)

b) Solve by matrix inversion method:  $3x - y + 2z = 13; 2x + y - z = 3; x + 3y - 5z = -8$

42. a) Resolve into partial fractions:  $\frac{x-2}{(x+2)(x-1)^2}$

(OR)

b) If  $A + B = 45^\circ$ , prove that  $(1 + \tan A)(1 + \tan B) = 2$  and hence deduce the value of  $\tan 22\frac{1}{2}$

43. a) Calculate the earliest start time, earliest finish time, latest start time and latest finish time of each activity of the project given below and determine the Critical path of the project and duration to complete the project.

| Activity           | 1-2 | 1-3 | 1-5 | 2-3 | 2-4 | 3-4 | 3-5 | 3-6 | 4-6 | 5-6 |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Duration (In week) | 8   | 7   | 12  | 4   | 10  | 3   | 15  | 10  | 7   | 4   |

(OR)



(OR)

b) Show that the middle term in the expansion of  $(1+x)^{2n}$  is  $\frac{1.3.5 \dots (2n-1)2^n x^n}{n!}$

44. a) Calculate Karl Pearson's coefficient of correlation from the following data:

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| X: | 6  | 8  | 12 | 15 | 18 | 20 | 24 | 28 | 31 |
| Y: | 10 | 12 | 15 | 15 | 18 | 25 | 22 | 26 | 28 |

(OR)

b) If  $y = a \cos mx + b \sin mx$ , then show that  $y_2 + m^2 y = 0$ .

45. a) Find the stationary values and stationary points for the function:  $f(x) = 2x^3 + 9x^2 + 12x + 1$ .

(OR)

b) Suppose the inter-industry flow of the product of two industries are given as under.

| Production Sector | Consumption sector |    | Domestic demand | Total output |
|-------------------|--------------------|----|-----------------|--------------|
|                   | X                  | Y  |                 |              |
| X                 | 30                 | 40 | 50              | 120          |
| Y                 | 20                 | 10 | 30              | 60           |

Determine the technology matrix and test Hawkin's -Simon conditions for the viability of the system. If the domestic demand changes to 80 and 40 units respectively, what should be the gross output of each sector in order to meet the new demands.

46. a) Which is better investment: 12% ₹ 20 shares at ₹ 16 (or) 15% ₹ 20 shares at ₹ 24.

(OR)

b) Solve the linear programming problem by graphical method.

Maximize  $Z = 6x_1 + 8x_2$  subject to constraints  $30x_1 + 20x_2 \leq 300$ ;  $5x_1 + 10x_2 \leq 110$ ; and  $x_1, x_2 \geq 0$ .

47. a) The heights (in cm.) of a group of fathers and sons are given below:

|                        |     |     |     |     |     |     |     |     |     |     |
|------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Heights of the fathers | 158 | 166 | 163 | 165 | 167 | 170 | 167 | 172 | 177 | 181 |
| Heights of the sons    | 162 | 158 | 167 | 170 | 160 | 180 | 170 | 175 | 172 | 175 |

Find the lines of regression and estimate the height of son when the height of the father is 164 cm.

(OR)

b) In a shooting test the probability of hitting the target are  $\frac{3}{4}$  for A,  $\frac{1}{2}$  for B and  $\frac{2}{3}$  for C. If all of them fire at the same target, calculate the probabilities:

(i) All the three hit the target

(ii) Only one of them hits the target

(iii) At least one of them hits the target.



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